

Sharp total variation bounds from spherical rearrangement

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Abstract

We study the sharp control of total variation by a higher derivative under homogeneous endpoint conditions. For every integer $n \geq 2$, we prove that the norm of $u \mapsto u'$ from $W_0^{n,2}(0,1)$ to $L^1(0,1)$ is twice the norm of point evaluation into $L^\infty(0,1)$. The extremizers of the two embeddings coincide and are symmetric about the midpoint. This proves the Hilbert-space case of a conjecture of Nazarov and Shcheglova. The main step is a rearrangement principle for a clamped Green operator after projection onto the zero-mean subspace. A Jacobi-polynomial representation identifies its quadratic form with a positive mixture of Poisson interactions on a sphere, where rearrangement reduces the optimization to a cap. We also prove the constant identity and coincidence of maximizing sets at the measure endpoint $p = 1$, using the geometry of extreme moment measures and unimodal splines. For each fixed derivative order, we further prove the full variation identity and midpoint-symmetric equality classification on an open interval of exponents around two. A dual optimality equation gives compactness through the highest derivative; polynomial sublevel bounds control the point-evaluation profile on both sides of the Hilbert exponent. At the measure endpoint in derivative order four, we also determine the exact sharp constant and unique symmetric extremizer by a cubic alternation argument and an explicit algebraic exclusion of asymmetric maximizers. The endpoint symmetry problem in higher orders and the full all-exponent conjecture remain open.

1. The sharp inequality

All functions and measures in this paper are real. For $1 \leq p \leq \infty$, we use the homogeneous norm $\|u^{(n)}\|_p$ on $W_0^{n,p}(0,1)$. The endpoint conditions are

$$u^{(j)}(0) = u^{(j)}(1) = 0, \quad 0 \leq j < n. \quad (1)$$

Let $C_{n,p}$ and $V_{n,p}$ denote the sharp constants in

$$\|u\|_\infty \leq C_{n,p} \|u^{(n)}\|_p, \quad \|u'\|_1 \leq V_{n,p} \|u^{(n)}\|_p. \quad (2)$$

The inequality $V_{n,p} \geq 2C_{n,p}$ follows from the variation of a function that vanishes at both endpoints. Nazarov and Shcheglova [NS, Conjecture 4.14] conjecture equality for all $n \geq 2$ and $1 \leq p \leq \infty$, together with coincidence and midpoint symmetry of the extremizers. Their Remark 4.15 records the previously known case $n = 2$.

Theorem 1. For every integer $n \geq 2$,

$$V_{n,2} = 2C_{n,2} = \frac{1}{2^{2n-2}(n-1)!\sqrt{2n-1}}. \quad (3)$$

The nonzero extremizers are precisely the scalar multiples of the Riesz representer $R_n(\cdot, 1/2)$ of evaluation at $1/2$ in $W_0^{n,2}(0,1)$. In particular, the maximizing sets in (2) coincide, and their elements are even about $1/2$.

The point-evaluation formula used here is prior work. Kalyabin [K, Theorem 2] gives, after affine rescaling,

$$\|\text{ev}_a\|^2 = \frac{[a(1-a)]^{2n-1}}{(2n-1)((n-1)!)^2}. \quad (4)$$

Our contribution is the upper bound for the variation norm and its equality classification, obtained through the rearrangement principle in Section 3. Best polynomial approximation already occurs in this embedding problem [GS]. The sharp $W_0^{m,2}$ to L^1 embedding of the function itself was proved by Hindov, Nitzan, Olsen and Rydhe [HNOR]; their theorem does not include the additional zero-mean constraint on a derivative that appears below.

2. A Jacobi representation of the clamped Green form

Fix $m \geq 1$, and set

$$w(x) = x^m(1-x)^m, \quad M = \int_0^1 w(x) dx = \frac{(m!)^2}{(2m+1)!}, \quad d\pi(x) = M^{-1}w(x) dx. \quad (5)$$

Let h_k be the orthonormal polynomials for π , with positive leading coefficients. These are normalized shifted Jacobi polynomials with parameters (m, m) ; in particular $h_0 = 1$. Put

$$v_k = wh_k, \quad \lambda_k = \frac{(k+2m)!}{k!}. \quad (6)$$

Lemma 2. The functions $v_k/\sqrt{M\lambda_k}$ form an orthonormal basis of $H = W_0^{m,2}(0,1)$ in its energy inner product. Moreover,

$$(-1)^m v_k^{(2m)} = \lambda_k h_k. \quad (7)$$

Proof. The left side of (7) has degree k . If q is a polynomial of degree less than k , integration by parts $2m$ times yields

$$\int_0^1 (wh_k)^{(2m)} wq = \int_0^1 wh_k (wq)^{(2m)} = 0. \quad (8)$$

Every boundary term vanishes because one of its two factors has fewer than m derivatives and retains an endpoint zero. Orthogonality therefore shows that the left side of (7) is proportional to h_k . Comparing leading coefficients proves (7). A further integration by parts gives

$$\langle v_k, v_\ell \rangle_H = M\lambda_k \delta_{k\ell}. \quad (9)$$

If $y \in H$ is orthogonal to every v_k , weak integration by parts gives $\int_0^1 y h_k = 0$ for every k . The h_k span the polynomials, which are dense in $C[0, 1]$, so $y = 0$. This proves completeness. $\square$

The mean functional $y \mapsto \int_0^1 y$ has Riesz representer $v_0/\lambda_0 = w/(2m)!$. Therefore

$$H_* := \left\{ y \in H : \int_0^1 y = 0 \right\} \quad (10)$$

is obtained by removing the zeroth basis vector. For a bounded real load g , Parseval gives the squared dual norm of $y \mapsto \int gy$ on H_* :

$$Q_m(g) = M \sum_{k \geq 1} \frac{\left| \int_0^1 g h_k d\pi \right|^2}{\lambda_k}. \quad (11)$$

Equivalently, if G_m is the clamped Green kernel and $w_m = w/(2m)!$, then

$$Q_m(g) = \iint g(x) G_m(x, t) g(t) dx dt - \frac{(\int g w_m)^2}{\int w_m}. \quad (12)$$

Formula (11) is sufficient for the proof and avoids any pointwise convergence issue for a Green series.

3. Spherical rearrangement of the load

Set $d = 2m + 2$ and let σ be normalized surface measure on S^d . The coordinate $x = (1 + \xi_{d+1})/2$ has law π . Thus

$$F(\xi) = g\left(\frac{1 + \xi_{d+1}}{2}\right) \quad (13)$$

identifies $L^2(\pi)$ with the zonal subspace of $L^2(S^d)$. Under this identification, h_k is the normalized zonal harmonic of degree k . This follows from the orthogonal decomposition of polynomials into spherical harmonics: a zonal polynomial of degree k orthogonal to all lower zonal degrees has only its degree- k component.

The spherical Poisson operator P_r , $0 < r < 1$, has eigenvalue r^k on degree- k harmonics and kernel

$$K_r(\xi \cdot \eta) = \frac{1 - r^2}{(1 - 2r\xi \cdot \eta + r^2)^{(d+1)/2}}. \quad (14)$$

We use normalized surface measure, as in [ABR, (1.15)]. The eigenvalue follows by harmonic extension of a homogeneous harmonic polynomial. The beta integral gives

$$\frac{1}{\lambda_k} = \frac{1}{(2m-1)!} \int_0^1 r^k (1-r)^{2m-1} dr. \quad (15)$$

Tonelli's theorem applied to the nonnegative spectral terms in (11) therefore gives

$$Q_m(g) = \frac{M}{(2m-1)!} \int_0^1 (1-r)^{2m-1} \left(\langle F, P_r F \rangle - \left(\int F d\sigma \right)^2 \right) dr. \quad (16)$$

The deletion of the constant harmonic is exactly the mean subtraction needed in (12).

Theorem 3. If $A \subset (0, 1)$ is measurable and $a \in [0, 1]$ satisfies $\pi(A) = \pi([0, a])$, then

$$Q_m(\mathbf{1}_A) \leq Q_m(\mathbf{1}_{[0,a]}). \quad (17)$$

If $0 < \pi(A) < 1$, equality holds only when A agrees, up to null sets, with an interval abutting one endpoint and having that π -measure.

Proof. Lift A to the zonal subset $E \subset S^d$ using (13). For fixed r , K_r is positive, bounded and strictly decreasing in geodesic distance. Spherical rearrangement [BH, Theorem 2] maximizes the self-interaction of $\mathbf{1}_E$ at fixed surface measure by a cap. Choose that cap about the negative coordinate axis. It corresponds to $[0, a]$. The mean term in (16) is unchanged, and integration proves (17).

For clarity, the equality hypotheses in [BH] are satisfied with two functions and integrand $(s, t) \mapsto st$. Its mixed finite difference is strictly positive; the kernel is positive and strictly decreasing; and the nontrivial indicators are nonconstant with finite interaction. Hence equality at fixed r forces E to be a cap up to rotation. If equality holds in (17), the nonnegative rearrangement deficit has zero integral in (16), so equality holds for almost every r , and in particular for one $r \in (0, 1)$. Thus E is a cap.

Its axis must be the original coordinate axis up to reversal. Indeed, zonal invariance makes the centroid of E parallel to that axis. A nontrivial cap has a nonzero centroid parallel to its own axis. Consequently A is an endpoint interval, as claimed. $\square$

The same argument proves a broader load rearrangement statement. For a bounded real g , let $g^\downarrow$ be its nonincreasing equimeasurable rearrangement relative to π . Adding a constant makes g nonnegative without changing Q_m . Applying [BH] in (16) gives

$$Q_m(g) \leq Q_m(g^\downarrow). \quad (18)$$

This is rearrangement of a load in a Green quadratic form. It does not assert that ordinary rearrangement preserves a higher derivative norm.

4. Proof of Theorem 1

Put $m = n - 1$. The map $u \mapsto y = u'$ is an isometric bijection from $W_0^{n,2}$ to H_* : integrating y recovers u , and the zero-mean condition supplies $u(1) = 0$.

Duality gives

$$V_{n,2}^2 = \sup_{|g| \leq 1} Q_m(g) = 4 \sup_A Q_m(\mathbf{1}_A). \quad (19)$$

To justify the second equality, write a bounded g with $|g| \leq 1$ as the average of $\text{sign}(g - s)$ over uniform $s \in [-1, 1]$. Convexity of the squared Hilbert dual norm bounds its value by the supremum over sign functions. Constants are annihilated in (11), giving $Q_m(2\mathbf{1}_A - 1) = 4Q_m(\mathbf{1}_A)$.

For an endpoint interval,

$$\int_0^a y(x) dx = u(a), \quad Q_m(\mathbf{1}_{[0,a]}) = \|\text{ev}_a\|^2. \quad (20)$$

Theorem 3, (4), and (19) now prove (3). In particular, the maximizing interval is a half-interval.

Suppose that $u \neq 0$ attains equality, and take $g = \text{sign}(u')$, choosing either sign where $u' = 0$. Both signs occur on sets of positive measure. Equality in the Hilbert dual bound and in Theorem 3 forces g

to be a half-interval sign function, up to its global sign and null sets. Its functional on H_* is $\pm 2u(1/2)$. Equality in Cauchy--Schwarz therefore makes u a scalar multiple of $R_n(\cdot, 1/2)$.

Conversely, this representer maximizes point evaluation. Its variation is at least twice its maximum and at most the bound just proved, so it also maximizes variation. Formula (4) has its unique maximum at $a = 1/2$, and uniqueness of a Hilbert-space Riesz representer gives precisely the same classification for the point-norm extremizers. Finally, reflection preserves the energy inner product and fixes evaluation at $1/2$, so uniqueness makes $R_n(\cdot, 1/2)$ reflection invariant. $\square$

5. The measure endpoint

Let $\mathcal{M}_n$ consist of real finite signed measures μ on $[0, 1]$ such that

$$\|\mu\|_{\text{TV}} \leq 1, \quad \int_0^1 t^j d\mu(t) = 0 \quad (0 \leq j < n). \quad (21)$$

Their compactly supported potentials are

$$U_\mu(x) = \int_0^1 \frac{(x-t)_+^{n-1}}{(n-1)!} d\mu(t). \quad (22)$$

The moments make U_μ vanish off $[0, 1]$, and $D^n U_\mu = \mu$. For $n \geq 2$, U_μ is continuous and absolutely continuous, with integrable first derivative. Endpoint atoms are allowed in this relaxed formulation.

Theorem 4. For every $n \geq 2$,

$$\sup_{\mu \in \mathcal{M}_n} \|U'_\mu\|_1 = 2 \sup_{\mu \in \mathcal{M}_n} \|U_\mu\|_\infty. \quad (23)$$

The maximizing measures for the two quantities coincide. The relaxed constants equal $V_{n,1}$ and $C_{n,1}$, respectively. No symmetry assertion at $p = 1$ is included.

Proof of the constant identity. Every extreme point of $\mathcal{M}_n$ has norm one and exactly $n+1$ support points. To prove the support bound, suppose μ has norm one and at least $n+2$ disjoint Borel sets of positive $|\mu|$ -measure. A nonzero function h constant on those sets can satisfy the n equations $\int t^j h d\mu = 0$ and the extra equation $\int h d|\mu| = 0$. Then $(1 \pm \varepsilon h)\mu$ are distinct norm-one admissible measures for small ε . Such a μ is not extreme. A nonzero measure of norm less than one is excluded by scalar perturbation. Zero is the midpoint of opposite nonzero moment measures, which exist on any $n+1$ distinct nodes, and is not extreme either. A nonzero measure on at most n points is excluded by the Vandermonde determinant.

On nodes $t_0 < \dots < t_n$, the moment nullspace is one-dimensional, with weights

$$b_i = \prod_{j \neq i} (t_i - t_j)^{-1}. \quad (24)$$

Thus every extreme measure is a normalized signed multiple of $\sum_i b_i \delta_{t_i}$. Its potential is a B-spline, up to scale and sign. More explicitly, the Genocchi--Hermite identity [dB, (52)] identifies

$$\rho(x) = n \sum_{i=0}^n b_i (t_i - x)_+^{n-1} = (-1)^n n! U_{\sum b_i \delta_{t_i}}(x) \quad (25)$$

with the density of $\sum_i t_i X_i$, where X is uniform on the standard n -simplex. After parallel sections of the simplex are identified by translation, their Minkowski interpolation is contained in the section at the interpolated coordinate. The classical Brunn--Minkowski inequality [GHW, (3), $p = 1$] therefore makes their volumes to the power $1/(n-1)$ concave. The coarea factor relating section volume to ρ is constant, so $\rho^{1/(n-1)}$ is concave and ρ is unimodal. It is positive between its end knots, continuous, and zero at those knots. Every extreme potential therefore satisfies

$$\|U'_\mu\|_1 = 2\|U_\mu\|_\infty. \quad (26)$$

Write $C = \sup_{\mathcal{M}_n} \|U_\mu\|_\infty$. The set $\mathcal{M}_n$ is weak-star compact. Moreover, $\mu \mapsto \|U'_\mu\|_1$ is weak-star lower semicontinuous: it is the supremum of the continuous linear functionals $\int H_g d\mu$, where $|g| \leq 1$ and

$$H_g(t) = \int_t^1 \frac{(x-t)^{n-2}}{(n-2)!} g(x) dx. \quad (27)$$

Its sublevel set at $2C$ is closed and convex and contains all extreme points by (26). Krein--Milman therefore gives the upper bound in (23). The elementary variation lower bound gives the reverse inequality. The height supremum is attained: on the bounded measure set, joint continuity of the kernels in (22) makes the potential map weak-star to uniform continuous. Consequently the variation supremum is attained as well.

Coincidence. Let μ maximize variation, choose $g = \text{sign}(U'_\mu)$, and let $\mathcal{F}$ be the compact exposed face on which $\int H_g d\nu = 2C$. Its extreme points are extreme in $\mathcal{M}_n$. For each such point e ,

$$2C = \int g U'_e \leq \|U'_e\|_1 = 2\|U_e\|_\infty \leq 2C. \quad (28)$$

Thus g agrees with the sign of U'_e wherever the latter is nonzero, and every U_e maximizes height. Its end knots must be 0 and 1: a potential supported on a shorter interval of length L can be stretched to $[0, 1]$, keeping its derivative-measure norm one and multiplying its height by $L^{1-n} > 1$.

All U_e have the same global sign, since opposite signs give opposite nonzero derivatives on a common interval near zero, contradicting their agreement with g . Change that common sign to positive. Concavity of $U_e^{1/(n-1)}$ makes its maximum set a closed interval, with strictly positive derivative almost everywhere before it and strictly negative derivative almost everywhere after it. Two disjoint maximum intervals would again force contradictory signs for g . The maximum intervals therefore intersect pairwise, and hence have a common point a_* . All $U_e(a_*) = C$. Evaluation is weak-star continuous, so Krein--Milman applied to $\mathcal{F}$ gives $U_\mu(a_*) = C$. Conversely, a height maximizer is a variation maximizer by the already proved constant identity.

Recovery of the ordinary constants. For an ordinary admissible u , the measure $u^{(n)} dx$ satisfies (21). Conversely, mollify the zero extension of U_μ with a nonnegative mollifier supported in $[-\varepsilon, \varepsilon]$, obtaining U_ε . With $a = 1 + 2\varepsilon$, set

$$u_\varepsilon(x) = a^{1-n} U_\varepsilon(ax - \varepsilon). \quad (29)$$

It is smooth, supported on $[0, 1]$, satisfies all endpoint conditions, and has $\|u_\varepsilon^{(n)}\|_1 \leq 1$. Its maximum and variation converge to those of U_μ , by uniform convergence of the potential mollification and L^1 convergence of its first derivative. This proves equality of the relaxed and ordinary sharp constants. The proof classifies coincidence of relaxed maximizers and does not assert attainment in the ordinary $W_0^{n,1}$ class. $\square$

6. An open interval of exponents

The Hilbert theorem also yields a neighborhood of exponents for each fixed derivative order. The interval is not asserted to be uniform in the order. In this section write $X_{n,p} = W_0^{n,p}(0,1)$ for the clamped space defined by (1), with norm $\|u^{(n)}\|_p$.

Theorem 5 (local continuation in the exponent). For every fixed integer $n \geq 3$, there exists $0 < \varepsilon_n < 1$ such that, if $|p - 2| < \varepsilon_n$, then

$$V_{n,p} = 2C_{n,p}. \quad (6.1)$$

All nonzero extremizers in the two inequalities are the scalar multiples of the unique positive norm-one extremizer for evaluation at $1/2$. They are symmetric about the midpoint and, after a choice of global sign, have strictly positive derivative on $(0, 1/2)$ and strictly negative derivative on $(1/2, 1)$.

The radius is allowed to depend on n and is not estimated here. The theorem does not reach all $1 < p < \infty$, give an interval uniform in n , or settle the remaining endpoint symmetry questions.

6.1. Dual representation and compactness of variation maximizers

Let $\mathcal{P}_{n-1}$ denote the polynomials of degree at most $n - 1$. Integration from the left endpoint identifies $X_{n,p}$ isometrically with

$$Z_{n,p} = \{f \in L^p(0,1) : \int_0^1 t^j f(t) dt = 0, \ 0 \leq j < n\}. \quad (6.2)$$

Indeed $u(x) = \int_0^x (x-t)^{n-1} f(t)/(n-1)! dt$; the right endpoint conditions are equivalent to the displayed moments. For bounded g , put

$$H_g(t) = \int_t^1 \frac{(x-t)^{n-2}}{(n-2)!} g(x) dx. \quad (6.3)$$

Fubini gives $\int g u' = \int H_g f$. If $q = p/(p-1)$, the norm of this functional on $Z_{n,p}$ equals

$$\min_{P \in \mathcal{P}_{n-1}} \|H_g - P\|_q. \quad (6.4)$$

Here is a direct justification without a quotient-duality assumption. A minimizing polynomial exists by finite-dimensional coercivity and is unique for $q > 1$. Its residual $R = H_g - P$ satisfies

$$\int_0^1 |R|^{q-2} R Q = 0 \quad (Q \in \mathcal{P}_{n-1}). \quad (6.5)$$

If $R \neq 0$, the function $f = \text{sign}(R)|R|/\|R\|_q^{q-1}$ has L^p norm one and belongs to (6.2). It realizes the bound in Hölder's inequality. The zero-residual case is immediate. This proves (6.4).

For every fixed $p > 1$, the variation supremum is attained. A norm-bounded sequence is bounded in $W^{n,p}$ by successive integration, and its derivatives through order $n - 1$ are uniformly bounded and equicontinuous, using Hölder's inequality for the highest integral. A subsequence converges uniformly through order $n - 1$, and the highest derivatives have a weakly convergent subsequence in L^p . The limit satisfies all endpoint conditions, has norm at most one, and retains the limiting variation. The supremum is positive, so a maximizer must have norm exactly one by scaling.

For a norm-one variation maximizer u , choose $g = \text{sign}(u')$, with any value in $[-1, 1]$ on the zero set. The functional in (6.4) is at most $V_{n,p}$, since $\int g v' \leq \|v'\|_1$. On u it equals $V_{n,p}$. Thus its residual satisfies $\|R\|_q = V_{n,p}$, and equality in Hölder gives

$$u^{(n)}(t) = \text{sign } R(t) \left| \frac{R(t)}{V_{n,p}} \right|^{q-1}. \quad (6.6)$$

This formula gives more compactness than the original Sobolev bound.

Lemma 6 (compactness of maximizers). Suppose $p_j \rightarrow 2$ and u_j is any sequence of norm-one variation maximizers in X_{n,p_j} . Then, after a subsequence and a choice of global signs, u_j converges in $C^n[0, 1]$ to the positive norm-one Hilbert maximizer u_* .

Proof. Restrict to $3/2 \leq p_j \leq 5/2$, so $5/3 \leq q_j \leq 3$. The functions H_{g_j} , with $|g_j| \leq 1$, are uniformly bounded and Lipschitz. For $n = 2$, this follows directly from $H'_g = -g$; for $n \geq 3$, differentiate (6.3) once and bound its remaining integral.

The minimizing polynomials satisfy $\|P_j\|_{q_j} \leq 2\|H_{g_j}\|_{q_j}$. Their coefficients are uniformly bounded. To verify the needed uniform norm equivalence, a contrary sequence of polynomials can be normalized to have coefficient norm one and L^{q_j} norm tending to zero. A coefficient-convergent subsequence would give a nonzero polynomial vanishing almost everywhere, a contradiction. Consequently both P_j and their derivatives are uniformly bounded, and the residuals R_j are uniformly bounded and Lipschitz.

The quantities V_{n,p_j} are bounded above by (6.4) with the zero polynomial. They are bounded away from zero by testing the one fixed nonzero function u_* , whose highest derivative is bounded. Formula (6.6) now makes $u_j^{(n)}$ uniformly bounded and equicontinuous. For example, signed power maps with exponent in $[2/3, 2]$ on a fixed bounded interval have a common Hölder modulus with exponent $2/3$, after enlarging the constant. Arzelà--Ascoli supplies a uniformly convergent subsequence of the highest derivatives. Integration and the left endpoint conditions give convergence in C^n to some u .

Uniform boundedness and $p_j \rightarrow 2$ imply $\int |u^{(n)}|^2 = \lim_j \int |u_j^{(n)}|^{p_j} = 1$. All right endpoint conditions pass to the limit. Also $\|u'\|_1 = \lim_j V_{n,p_j}$. Testing $u_*/\|u_*\|_{p_j}$ gives $\liminf_j V_{n,p_j} \geq V_{n,2}$, while the Hilbert inequality applied to the limit gives the reverse bound. Thus u is a Hilbert maximizer. Theorem 1 identifies it with u_* or $-u_*$, proving the lemma. $\square$

6.2. Persistence of a single peak

We give the details that weak or uniform convergence alone would miss. The Hilbert midpoint representer, up to a positive factor, can be written

$$U(x) = F(|1 - 2x|), \quad F(z) = \int_z^1 (t^2 - z^2)^{n-1} dt, \quad 0 \leq z \leq 1. \quad (6.7)$$

To check this representation, expand the integrand. The resulting expression for F is an even polynomial of degree at most $2n - 2$, plus a multiple of z^{2n-1} . Thus (6.7) is polynomial of degree at most $2n - 1$ on each half-interval and has matching derivatives through order $2n - 2$ at the midpoint. Near $z = 1$, writing $t = z + (1 - z)s$ shows that $F(z)$ is a positive constant times $(1 - z)^n$, to leading order. Therefore all endpoint derivatives below order n vanish. The jump of the derivative of order $2n - 1$ is nonzero. The weak derivative $(-1)^n D^{2n} U$ is a scalar multiple of the midpoint delta. Testing against U shows this scalar is positive because $\int |U^{(n)}|^2 > 0$ and $U(1/2) > 0$. Uniqueness of the Riesz representer proves the claim up to positive normalization.

For $n \geq 2$, differentiation gives

$$F'(z) = -2(n-1)z \int_z^1 (t^2 - z^2)^{n-2} dt < 0 \quad (0 < z < 1). \quad (6.8)$$

Also $F''(0) = -2(n-1)/(2n-3) < 0$. Hence $u'_* > 0$ on $(0, 1/2)$, $u'_* < 0$ on $(1/2, 1)$, and $u''_*(1/2) < 0$. The endpoint expansion gives $u_*^{(n)}(0) > 0$ and $(-1)^n u_*^{(n)}(1) > 0$.

These properties imply that every sufficiently close C^n perturbation satisfying the same clamped endpoint conditions has exactly one interior critical point, a strict maximum. Indeed, near zero its highest derivative remains positive and

$$u'(x) = \int_0^x \frac{(x-t)^{n-2}}{(n-2)!} u^{(n)}(t) dt > 0. \quad (6.9)$$

The reflected argument gives negativity near one. On compact intervals away from the endpoints and midpoint the strict derivative signs persist. On a small interval about the midpoint, the second derivative stays negative, so there is precisely one zero of the first derivative, bracketed by its signs on either side.

Lemma 6 therefore implies that for all p sufficiently close to two, every norm-one variation maximizer is, up to sign, positive and strictly unimodal. Such a maximizer has variation exactly twice its height. Since every admissible function has variation at least twice its height, this proves (6.1). It also proves coincidence of maximizing sets: a variation maximizer has height $C_{n,p}$, and every height maximizer has variation between $2C_{n,p}$ and $V_{n,p}$.

The only conclusion still needed for the local theorem is midpoint symmetry. Single-peak geometry by itself does not supply it.

6.3. Smoothness of the point-evaluation profile near two

Let $m = n - 1 \geq 2$, and define

$$K_a(t) = \frac{(a-t)_+^m}{m!}, \quad J(q, a) = \min_{P \in \mathcal{P}_m} \int_0^1 |K_a - P|^q dt. \quad (6.10)$$

By the same moment argument as (6.4), the evaluation norm is $J(q, a)^{1/q}$. We need uniform continuity on the whole interval in a , and twice continuous differentiability in a near $(q, a) = (2, 1/2)$.

First, minimizing polynomials have coefficients bounded uniformly for q in a compact subset of $(1, \infty)$ and $a \in [0, 1]$, by the argument used above. On this bounded coefficient set the integrands in (6.10) vary continuously with (q, a, P) , uniformly in t . Minimizing over that compact set proves continuity of J , including uniform convergence $J(q, \cdot) \rightarrow J(2, \cdot)$ as $q \rightarrow 2$. Strict convexity gives a unique minimizing polynomial; compactness and uniqueness prove continuous dependence of its coefficients.

The following elementary sublevel estimate handles exponents just below two. If P is a polynomial of degree at most m on an interval I of length bounded above and below, and $\|P\|_{L^\infty(I)} \geq b > 0$, then

$$|\{t \in I : |P(t)| \leq h\}| \leq C(m, b, I) h^{1/m} \quad (0 < h < 1). \quad (6.11)$$

The constant is uniform for intervals in the stated length range. To prove this, let the sublevel set have length $\ell > 0$. Divide its mass into sufficiently many equal portions and choose $m + 1$ points from alternate portions. They can be chosen in the sublevel set with consecutive separations bounded

below by a fixed multiple of $\ell/(m+1)$: coordinate separation is at least the measure of the intervening portion. Lagrange interpolation at these points gives $\|P\|_\infty \leq C_m h \ell^{-m}$, with the interval-length factor absorbed into the constant. Rearrangement gives (6.11); approximation of the chosen points handles essential rather than attained sublevel membership.

Consequently, for $0 < s < 1/m$, integration of distribution functions gives

$$\int_{\{|P|<h\}} |P|^{-s} dt \leq C h^{1/m-s}. \quad (6.12)$$

For example, split the region into dyadic bands $2^{-k-1}h \leq |P| < 2^{-k}h$, apply (6.11) to each, and sum the convergent geometric series. The estimates also show uniform integrability of these inverse powers.

At $(q, a) = (2, 1/2)$, the minimizing residual in (6.10) is a nonzero polynomial on each of the two half-intervals. If it were identically zero on one half, its orthogonality to constants would fail on the other: the remaining residual would be a nonzero constant multiple of the nonnegative truncated power. Continuity of the coefficients therefore supplies a uniform positive lower bound for the sup norm of each polynomial piece on its own interval when (q, a) and the polynomial coefficients are sufficiently close to this point. The intervals' lengths also stay bounded away from zero. Thus (6.11)–(6.12) apply uniformly to both pieces.

Write $P_c(t) = \sum_{j=0}^m c_j t^j$ and $R = K_a - P_c$. The normal equations are

$$\mathcal{N}_j(q, a, c) = \int_0^1 |R|^{q-2} R t^j dt = 0, \quad 0 \leq j \leq m. \quad (6.13)$$

Take a small open neighborhood with $q > 2 - 1/(2m)$. The equations are continuously differentiable in (q, a, c) there. Their derivatives in (a, c) are obtained by multiplying the variation of R by $(q-1)|R|^{q-2}$, and the derivative in q has integrand $|R|^{q-2} R \log |R| t^j$, with value zero at $R = 0$. These assertions follow first away from a small sublevel set of R , by ordinary differentiation. The uniform bound (6.12) makes the omitted derivatives uniformly integrable and arbitrarily small as the sublevel threshold tends to zero. The logarithmic derivative is bounded by a constant times a small positive power of $|R|$ near zero. This also proves continuity of the derivatives by splitting into the same two regions. The moving interface $t = a$ causes no additional term, since K_a and its first a -derivative are continuous there for $m \geq 2$.

At the base point, the coefficient derivative of (6.13) is the negative definite Gram matrix $-\left[\int_0^1 t^{i+j} dt\right]_{i,j=0}^m$. The finite-dimensional implicit function theorem yields a continuously differentiable coefficient map $c(q, a)$. It is the minimizing polynomial map by strict convexity and continuity already established.

The envelope derivative and its second derivative are

$$\begin{aligned} J_a &= q \int |R|^{q-2} R K_a^{[1]}, \\ J_{aa} &= q(q-1) \int |R|^{q-2} (K_a^{[1]} - P_{c_a}) K_a^{[1]} + q \int |R|^{q-2} R K_a^{[2]}, \end{aligned} \quad (6.14)$$

where $K_a^{[1]} = (a-t)_+^{m-1}/(m-1)!$ and $K_a^{[2]} = (a-t)_+^{m-2}/(m-2)!$. For $m = 2$, the last expression is the indicator of $t < a$, with its value at $t = a$ immaterial. The coefficient term in the first derivative vanishes by (6.13). The second derivative exists and is continuous by (6.12), boundedness of these kernels, and dominated convergence away from the moving interface. In particular no claim that all residual roots are simple is needed.

At $q = 2$, the credited point-evaluation formula (4) gives

$$J(2, a) = \frac{[a(1 - a)]^{2n-1}}{(2n - 1)((n - 1)!)^2}. \quad (6.15)$$

It has a unique maximum at $a = 1/2$, with strictly negative second derivative there. By continuity of (6.14), it follows that $J(q, \cdot)$ is strictly concave on one fixed neighborhood of $1/2$, for all q close enough to two. Uniform convergence on $[0, 1]$ forces every global maximum into that neighborhood. Reflection of the original evaluation problem gives $J(q, a) = J(q, 1 - a)$. Strict concavity and this symmetry show that $1/2$ is the unique global maximizing point.

6.4. Completion of the proof

Shrink ε_n so that both the variation-maximizer argument and the evaluation-profile argument apply. We have established the constant identity and coincidence of maximizing sets. Every height maximizer must realize evaluation at its absolute maximum, and Section 6.3 forces that point to be $1/2$. The norm-one positive evaluation extremizer is unique by equality in Hölder in (6.4), or by strict convexity of the unit ball of L^p . Reflection preserves the norm and evaluation at $1/2$, so uniqueness makes this function midpoint symmetric. Section 6.2 already gives its strict derivative signs and its unique peak. This proves Theorem 5. $\square$

6.5. The unresolved global step

Compactness away from the endpoints of the exponent range. The argument in Lemma 6 gives the following more general statement. If $p_j \rightarrow p_0 \in (1, \infty)$ and u_j are norm-one variation maximizers, a subsequence converges in C^n to a norm-one variation maximizer at p_0 , and $V_{n,p_j} \rightarrow V_{n,p_0}$. To see this, keep p_j in any fixed compact subinterval of $(1, \infty)$; the exponents $q_j - 1$ then have a positive lower bound, so (6.6) still gives a common Hölder modulus. The limit has norm one at p_0 . For the lower bound on the limiting optimal value, test a maximizer at p_0 ; (6.6) makes its highest derivative bounded, so it is admissible at all nearby exponents and its varying norms converge. The upper bound follows by applying the p_0 inequality to the limit. This proves the statement without assuming uniqueness or symmetry at p_0 .

Thus a local continuation criterion at another exponent is available: if all its normalized variation maximizers are scalar-sign copies of one single-peak function with nonzero endpoint highest derivatives and a nondegenerate peak, and its point-evaluation profile has a unique midpoint maximum with a continuously persisting negative second derivative, the same argument proves the conjecture in a neighborhood of that exponent. These hypotheses must be established there; they cannot be imported from the case $p = 2$.

The strongest compactness assertion here is C^n convergence of *variation maximizers*, established from their dual optimality equation. It is not asserted for arbitrary normalized Sobolev sequences. This distinction is necessary to preserve the derivative signs near the clamped endpoints.

The point-profile calculation uses finite-dimensional best approximation only after proving uniform control of possible multiple zeros in its polynomial residual. Omitting (6.11)–(6.12) would leave the two-sided neighborhood in p unjustified, since $q < 2$ creates negative powers at residual zeros.

This argument does not prove that the set of valid exponents is closed. At a hypothetical first exponent where the result ceases to hold, compactness of the old single-peak branch does not exclude a different multiple-peak global maximizer with the same value. Nor does local strict concavity of

the point-evaluation profile exclude a remote competing maximum at other exponents. These are substantive global obstacles, not consequences of the implicit function theorem.

The remaining task is to exclude such competing branches or to find a counterexample. Theorem 5 supplies a local result, not the full conjecture. No numerical experiment is used in its proof.

7. Complete symmetry at the fourth-order measure endpoint

The remaining endpoint symmetry problem can be settled in order four by a finite alternation argument. The theorem below concerns the measure class (21); its constants also apply to the ordinary clamped Sobolev space by Theorem 4.

Theorem 8 (fourth-order endpoint). One has

$$C_{4,1} = \frac{1}{1296}, \quad V_{4,1} = \frac{1}{648}. \quad (7.1)$$

The norm-one relaxed extremizers for height and variation are exactly μ_* and $-\mu_*$, where

$$\mu_* = \frac{1}{9}(\delta_0 + \delta_1) - \frac{1}{4}(\delta_{1/6} + \delta_{5/6}) + \frac{5}{18}\delta_{1/2}. \quad (7.2)$$

Their potentials are even about $1/2$. No nonzero extremizer exists in the ordinary $W_0^{4,1}$ class. Thus the full constant, coincidence and symmetry assertions in [NS, Conjecture 4.14] hold for $n = 4$, $p = 1$, with the endpoint understood in the measure sense.

The theorem does not assert symmetry at $p = 1$ for every derivative order. The point estimates for orders two and three are already included in [NS, Theorem 4.5]; those cases are not new here.

7.1. Extreme measures and a dual cubic

Let C be the maximal relaxed height. It is positive and attained by Section 5. Choose a maximizing point and sign. The corresponding evaluation functional exposes a nonempty compact face of $\mathcal{M}_4$. An extreme point of that face is extreme in $\mathcal{M}_4$ and still maximizes height. By the moment-circuit argument in Theorem 4, after choosing its positive potential it has nodes

$$0 < a < b < c < 1, \quad (t_0, t_1, t_2, t_3, t_4) = (0, a, b, c, 1), \quad (7.3)$$

$$\mu = \frac{1}{D} \sum_{i=0}^4 \beta_i \delta_{t_i}, \quad \beta_i = \prod_{j \neq i} (t_i - t_j)^{-1}, \quad D = \sum_i |\beta_i|.$$

The end nodes are 0 and 1 because stretching a shorter support, with derivative-measure norm fixed, multiplies height by the inverse cube of its length. The signs of the weights in (7.3) are $+, -, +, -, +$.

Write U for this positive potential and let x maximize it. Before a it is a positive multiple of x^3 ; after c it is a positive multiple of $(1 - x)^3$. Its derivatives at a and c have strict opposite signs. Hence $a < x < c$. Reflecting if necessary, assume

$$a < x \leq b < c < 1. \quad (7.4)$$

Put $K_x(t) = (x - t)_+^3/6$. There is a polynomial P of degree at most three such that

$$R(t) = K_x(t) - P(t), \quad \|R\|_\infty = C, \quad (R(0), R(a), R(b), R(c), R(1)) = (C, -C, C, -C, C). \quad (7.5)$$

Here are the duality details. The dual of $C[0, 1]/\mathcal{P}_3$ is the space of finite measures annihilating $\mathcal{P}_3$, with total-variation norm. Hahn-Banach and the Riesz representation theorem therefore identify the distance of K_x to $\mathcal{P}_3$ with the norm of its evaluation functional on $\mathcal{M}_4$. That norm equals C , because the chosen measure attains the global height. A best polynomial exists by finite-dimensional coercivity. Equality in $\int R d\mu \leq \|R\|_\infty \|\mu\|_{\text{TV}}$ forces the five values in (7.5). Since R is continuously differentiable, its derivative vanishes at a, b, c .

Let $R_+ = -P$, so that $R = R_+$ on $[x, 1]$. The cubic R_+ has a maximum at b and a minimum at c . Consequently, for some $k > 0$,

$$R'_+(t) = k(t - b)(t - c). \quad (7.6)$$

Equality of its values at b and 1 gives, on putting $h = c - b$,

$$1 - b = \frac{3}{2}h, \quad b = 1 - \frac{3}{2}h, \quad c = 1 - \frac{1}{2}h. \quad (7.7)$$

Indeed $\int_b^1 (t - b)(t - c)dt = (1 - b)^2((1 - b)/3 - h/2) = 0$.

7.2. The remaining algebraic constraints

On the left of x , $R(t) = R_+(t) + (x - t)^3/6$. The equations $R'(a) = 0$ and $R(a) = R(c)$ yield

$$(x - a)^2 = 2k(b - a)(c - a), \quad (x - a)^3 = k(c - a)^2(2(c - a) - 3h). \quad (7.8)$$

Set $v = (c - a)/h > 1$. Dividing the two equations, and using $0 < x - a \leq b - a$, gives

$$\frac{3}{2} < v \leq 2, \quad a = 1 - h\left(v + \frac{1}{2}\right), \quad x = 1 - \frac{h(2v - 1)}{2(v - 1)}. \quad (7.9)$$

For clarity, $(x - a)/h = v(2v - 3)/(2(v - 1))$. Its positivity gives $v > 3/2$, and its upper bound $v - 1$ gives $v \leq 2$.

The potential itself is stationary at x . Only the nodes $0, a$ contribute to its first derivative under (7.4), so $U'(x) = 0$ reads

$$(x - a)^2 = x^2 \frac{(b - a)(c - a)(1 - a)}{bc}. \quad (7.10)$$

Combining this with (7.8) gives $k = x^2(1 - a)/(2bc)$. Also $R(0) = R(1)$ implies

$$\frac{x^3}{6} = k \int_0^1 (t - b)(t - c)dt = \frac{kb^2}{3}, \quad x = \frac{b(1 - a)}{c}. \quad (7.11)$$

The integral equals $b^2/3$ by (7.7). All divided quantities are positive.

Substitute (7.7) and (7.9) into (7.10)-(7.11). The resulting two quadratic equations in h are

$$Ah^2 + Bh + C_0 = 0, \quad D_0h^2 + Eh + F = 0, \quad (7.12)$$

where

$$\begin{aligned} A &= 6v^2 - v - 4, & B &= -4v^2 - 4v + 6, \\ C_0 &= 4v - 4, & D_0 &= 2(v-1)(2v-1)^2, \\ E &= -16v^3 + 42v^2 - 31v + 4, & F &= 2v(2v-3)^2. \end{aligned} \quad (7.13)$$

One can check the second equation without elimination software: after substitution, the difference between the two sides of (7.10), multiplied by bc , is

$$(x-a)^2bc - x^2(b-a)(c-a)(1-a) = \frac{ah^2v}{8(v-1)^2}(D_0h^2 + Eh + F). \quad (7.14)$$

The first equation is $4(v-1)(xc - b(1-a)) = 0$. These identities justify (7.12) with no loss from a vanishing factor.

Two quadratics in (7.12) with a common root must satisfy

$$0 = (AF - C_0D_0)^2 - (AE - BD_0)(BF - C_0E) = 16(v-2)^2(v-1)(2v+1)\Pi(v), \quad (7.15)$$

where direct expansion gives

$$\Pi(v) = 8v^6 - 24v^5 - 6v^4 + 64v^3 - 48v^2 + 9v - 2. \quad (7.16)$$

To verify the first equality directly, put $L = AE - BD_0$, $M = AF - C_0D_0$ and $N = BF - C_0E$. Subtracting multiples of the two equations gives $Lh + M = 0$ and $h(Mh + N) = 0$. Since $h > 0$, these imply $M^2 - LN = 0$. The factorization in the second equality is an ordinary polynomial identity.

There is a short sign check for its last factor. For $z = v - 3/2 \in [0, 1/2]$,

$$\begin{aligned} \Pi(3/2 + z) &= 8z^6 + 48z^5 + 84z^4 + 28z^3 - \frac{87}{2}z^2 - 27z - 2 \\ &\leq -2z^2 - 27z - 2 < 0. \end{aligned} \quad (7.17)$$

Indeed the four positive terms are at most $(1/2 + 6 + 21 + 14)z^2 = (83/2)z^2$. Since $v > 3/2$, equation (7.15) forces $v = 2$. Both equations in (7.12) then reduce to $2(3h-2)(3h-1) = 0$. The root $h = 2/3$ makes $a < 0$ and is inadmissible. The other gives

$$(a, b, c, x) = \left(\frac{1}{6}, \frac{1}{2}, \frac{5}{6}, \frac{1}{2}\right). \quad (7.18)$$

Thus every globally maximizing extreme measure has precisely these nodes. This conclusion concerns maximizers directly; it does not require a knot-reflection inequality for every configuration.

7.3. Constants and all equality cases

The weights in (7.3), normalized to total variation one at (7.18), are exactly (7.2). Their potential is

$$U_*(x) = \frac{1}{6} \left[\frac{x_+^3 + (x-1)_+^3}{9} - \frac{(x-1/6)_+^3 + (x-5/6)_+^3}{4} + \frac{5}{18}(x-1/2)_+^3 \right]. \quad (7.19)$$

It is symmetric and unimodal by Section 5. Substitution at $x = 1/2$ gives $U_*(1/2) = 1/1296$. The extreme maximizing measure exists, and (7.18) identifies it, so this value is the unrestricted sharp height, not merely the maximum within symmetric configurations. Theorem 4 then gives the variation constant $1/648$.

For an arbitrary norm-one height maximizer, expose a face by evaluation at one of its maximum-height points, with the appropriate sign. Every extreme point of that face is a height-maximizing extreme measure and therefore equals μ_* or $-\mu_*$. Only one of these two signs can belong to the chosen face, because the exposed value is positive. Krein-Milman makes the face a singleton. This proves the stated equality classification, and Theorem 4 transfers it to variation. An ordinary Sobolev extremizer would have an absolutely continuous fourth derivative, whereas (7.2) is atomic. Hence ordinary attainment is impossible. The approximation argument in Theorem 4 still recovers both sharp constants. $\square$

8. Remaining questions

The full all-exponent form of [NS, Conjecture 4.14] remains unresolved. For each order, Theorem 5 proves an open interval about two, while Theorem 4 proves the constant identity and coincidence at the measure endpoint. Theorem 8 completes the symmetry and knot classification for the fourth-order measure endpoint. The endpoint symmetry problem for orders at least five is still open here. Further continuation in the exponent requires excluding competing maximizers or finding a counterexample; the compactness argument alone does not do that. The argument in Section 7 classifies global maximizers directly and does not establish a reflection inequality for every knot configuration.

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